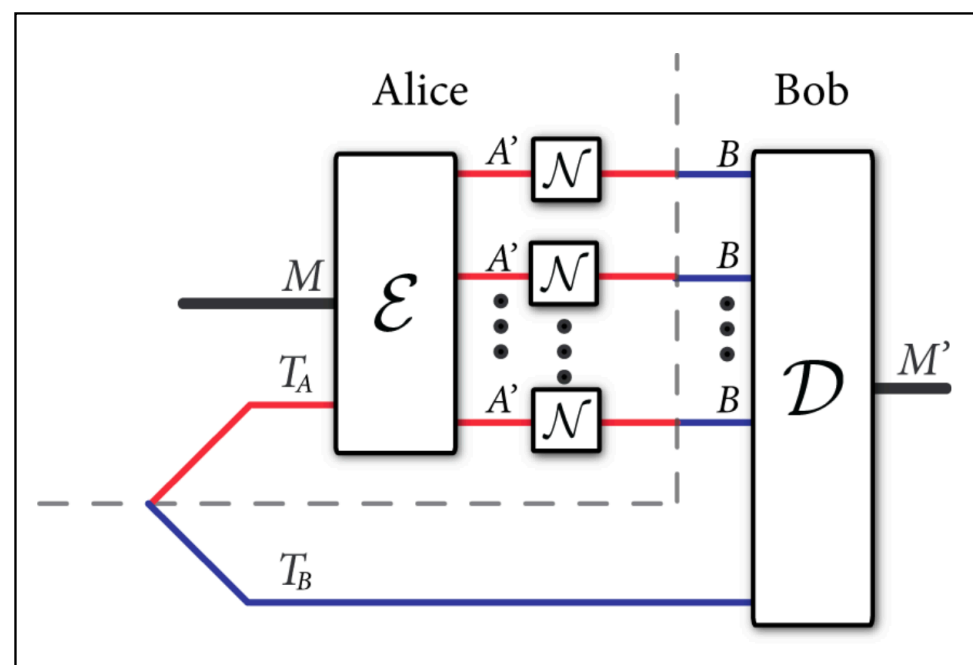


Entanglement Assisted Classical Capacity



- Super-dense coding: Entanglement can double the classical capacity of a noiseless qubit channel.
- Generalisation of Superdense Coding to arbitrary channels: Entanglement Assisted Classical Capacity
- Alice & Bob share copies of pure entangled state: $\Psi_{T_A T_B}$
- Entanglement Assisted Capacity (Maximum of Quantum Mutual Information) :
- $C_E(\mathcal{N}) = I(\mathcal{N}) = \max I(A; B)_\rho$ where $\varphi_{AA'}$ is a pure bipartite state and $\rho_{AB} = \mathcal{N}_{A \rightarrow B}(\varphi_{AA'})$
- A' is the system transmitted through the channel (red wire in the figure), while A is an untouched reference/purifying system used to quantify how much information the channel preserves.





Entanglement Assisted Classical Capacity Examples

- Ex-1: Identity (Noiseless) Qubit Channel: $\mathcal{N}(\rho) = \rho$
- Ex-2: Completely Depolarising Qubit Channel: $\mathcal{N}(\rho) = \frac{I}{2}$
- Ex-3: Bit Flip Channel: $\mathcal{N}(\rho) = (1 - p)\rho + pX\rho X$

Entanglement Assisted Capacity of Identity (Noiseless) Qubit Channel: $\mathcal{N}(\rho) = \rho$



Ex-1 Noiseless Channel : $\mathcal{N}(\rho) = \rho$

Alice & Bob share maximally entangled state:

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Alice sends one Qubit through the channel.

$$C_E = I(A;B) = S(A) + S(B) - S(AB).$$

$$S(AB) = 0 \quad (\text{Bell state is pure}).$$

$$\rho_A \stackrel{?}{=} \text{Tr}_B(\rho_{AB}) = I/2 \Rightarrow S(A) \stackrel{?}{=} 1.$$

$$h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p) \quad \text{if } p = 1/2$$

$$S(A) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1 \text{ bit}$$

$$S(B) = ?$$

$$\rho_B = ? \quad \text{Tr}_A(\rho_{AB}) = I/2$$

$$S(B) = 1$$

$$C_E = 1 + 1 - 0 = 2 \text{ bit/ch use} \rightarrow \text{Valid Superdense Coding.}$$

Holevo Capacity: 1 bit/ch use $\uparrow 2x$.

Entanglement Assisted Capacity of Completely Depolarising Qubit Channel: $\mathcal{N}(\rho) = \frac{I}{2}$



Ex-2 Completely depolarizing channel $\mathcal{N}(\rho) = \frac{I}{2}$.

Step-1 Sending Half of the Bell pairs $|\Phi^+\rangle$ (Initial)

After the depolarizing channel?

$$\rho_A = \rho_B = \frac{I}{2}$$

$$\rho_{AB} = \rho_A \otimes \rho_B$$

$$\rho_{AB} = \frac{I}{2} \otimes \frac{I}{2}$$

$$C_E = S(A) + S(B) - S(AB)$$

$$= 1 + 1 - S(AB)$$

$$= 2 - 2 = 0$$

$$= \frac{1}{4} I_{4 \times 4}$$

$$\sum_{i=1}^4 -\frac{1}{4} \log \frac{1}{4} = 2.$$

$$S(AB) = 2$$



Entanglement Assisted Capacity of Bit Flip Channel: $\mathcal{N}(\rho) = (1-p)\rho + pX\rho X$

Ex-3 Bit Flip Channel: $\mathcal{N}(\rho) = (1-p)\rho + pX\rho X$
 $C_E(\mathcal{N}) = \max_{\phi^{AA'}} I(A'; B)$
 For Pauli Channels (Bit flip channel); the maximizing input is the maximally entangled Bell state $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Alice sends the "second" Qubit through the channel.
 $|\phi^+\rangle \rightarrow 1/p$ state
 If No Bit flip occurs then, $|\phi^+\rangle \rightarrow |\phi^+\rangle (1-p)$
 If Bit flip occurs then, $|\phi^+\rangle \rightarrow |\psi^+\rangle (p)$
 $\frac{1}{\sqrt{2}} (|00\rangle_{AB} + |11\rangle_{AB}) \rightarrow \frac{1}{\sqrt{2}} (|11\rangle + |01\rangle) = |\psi^+\rangle$
 o/p Density operator $\rho_{AB} = (1-p) |\phi^+\rangle\langle\phi^+| + p |\psi^+\rangle\langle\psi^+|$

$C_E = S(A) + S(B) - S(AB)$
 $\rho_A = I/2 \Rightarrow S(A) = 1$
 $\rho_B = I/2 \Rightarrow S(B) = 1$
 $I/2$
 For All Bell States Partial Trace gives maximally mixed state. i.e. $\rho_A = \rho_B = I/2$
 $\rho_A = \text{Tr}_B(\rho_{AB}) = (1-p) \text{Tr}_B(|\phi^+\rangle\langle\phi^+|) + p \text{Tr}_B(|\psi^+\rangle\langle\psi^+|)$
 $= (1-p) \frac{I}{2} + p \frac{I}{2}$
 $\rho_A = I/2$ Similarly $\rho_B = I/2$
 $S(A) = S(B) = 1$
 $\rho_{AB} \rightarrow$ For 2 Qubit, Bell States form a valid Bases.
 $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\} \rightarrow$ Standard Bases for 2 Qubit.

Entanglement Assisted Capacity of Bit Flip Channel: $\mathcal{N}(\rho) = (1-p)\rho + pX\rho X$



Bell Basis $\rightarrow \{ |\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle \}$

Find the eigenvalues of ρ_{AB} in the Bell Basis.

$$\rho_{AB} = \sum_{i=1}^4 \lambda_i |\phi^i\rangle\langle\phi^i| \rightarrow \begin{aligned} \lambda_1 &= p \\ \lambda_2 &= 1-p \\ \lambda_3 &= 0 \\ \lambda_4 &= 0 \end{aligned}$$

$$|00\rangle = \frac{1}{\sqrt{2}} (|\phi^+\rangle + |\phi^-\rangle)$$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle), \quad |\phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\begin{aligned} S(\rho_{AB}) &= - \sum \lambda_i \log \lambda_i = -p \log_2(p) - (1-p) \log_2(1-p) \\ &= h_2(p) \end{aligned}$$

$$\begin{aligned} C_E &= S(A) + S(B) - S(\rho_{AB}) \\ &= 1 + 1 - h_2(p) = 2 - h_2(p) \end{aligned}$$

Holevo Capacity of Bit flip Channel = $1 - h_2(p)$

